FILMORA

FRONTEND ARCHITECTURE & REQUIREMENTS SPECIFICATION

Project Name: FILMORA
Project Type: Free Online Movie Streaming Platform
Document Type: Frontend Technical Specification
Version: 1.0
Status: Proposed Design
Frontend Technologies: HTML5, CSS3 and JavaScript
Backend Integration: Node.js + Express.js
Database: PostgreSQL
Design Style: Dark Cinematic
Target Devices: Desktop, Tablet and Mobile

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1. INTRODUCTION

The FILMORA frontend is the visual and interactive part of the movie streaming platform.

It provides the interface through which registered users interact with the platform, discover movies, view movie information and watch authorized full-length movies.

The frontend will communicate with the backend through REST APIs.

The frontend will be developed using HTML5, CSS3 and JavaScript.

1.1 Main objectives

- Create a professional movie streaming interface.
- Provide easy website navigation.
- Allow users to discover movies.
- Provide attractive movie posters and descriptions.
- Support full-screen video playback.
- Provide user registration and login.
- Support mobile and desktop devices.

- Provide an administrator interface.
- Maintain consistent design across all pages.

2. FRONTEND OBJECTIVES

The frontend must satisfy the following requirements:

1. Simple and understandable navigation.
2. Professional cinematic appearance.
3. Responsive layouts.
4. Fast page loading.
5. Interactive movie cards.
6. Functional search.
7. User-friendly forms.
8. Secure communication with the backend.
9. Accessible interface components.
10. Consistent colors, typography and spacing.

3. FRONTEND TECHNOLOGY STACK

Technology	Purpose
HTML5	Website structure
CSS3	Styling and layout
JavaScript	Interactivity
Fetch API	Backend communication
HTML5 Video	Video playback
HLS.js	HLS playback support where required
Google Fonts	Typography
SVG	Icons and graphics

The initial project will use vanilla JavaScript without requiring a frontend framework.

4. WEBSITE DESIGN REQUIREMENTS

FILMORA will use a dark cinematic design inspired by modern movie streaming platforms.

4.1 Design characteristics

- Dark backgrounds.
- Red accent colors.
- Large movie posters.
- Cinematic featured banners.
- Horizontal movie collections.
- Smooth hover effects.
- Rounded movie cards.
- Clear navigation.
- Attractive typography.
- Minimal unnecessary visual elements.

4.2 General interface layout

The main website pages will generally contain:

1. Navigation bar.
2. Main content section.
3. Movie collections or page-specific content.
4. Footer.

The navigation bar will contain:

- FILMORA logo.
- Home.
- Movies.
- Genres.
- Search.
- User profile.
- Logout option.

5. COLOR SCHEME AND TYPOGRAPHY

5.1 Main color palette

Color	Hexadecimal Usage

Color	Hexadecimal	Usage
Background	#09090F	Main background
Secondary Background	#12121B	Content sections
Card Background	#1B1B27	Movie cards
Primary Red	#E50935	Main accent
Hover Red	#B80729	Hover effects
White	#FFFFFF	Main text
Light Gray	#CCCCCC	Secondary text
Muted Gray	#888888	Metadata

5.2 Typography

Proposed fonts:

- Inter for general website text.
- Poppins for headings.
- Arial as a fallback font.

Typography must remain readable across all supported devices.

6. FRONTEND ARCHITECTURE

The frontend will be organized into reusable pages and components.

Main interface components:

- Navigation bar.
- Footer.
- Movie card.
- Movie carousel.
- Search bar.
- Genre selector.
- Movie information panel.
- Video player.
- Login form.

- Registration form.
- User profile menu.
- Notification component.
- Confirmation dialog.
- Loading indicator.
- Error message component.

Reusable components will maintain a consistent appearance across the platform.

7. FRONTEND FOLDER STRUCTURE

Proposed project structure:

FILMORA/

frontend/

- index.html
- pages/
    - login.html
    - register.html
    - forgot-password.html
    - movies.html
    - movie-details.html
    - watch.html
    - profile.html
    - watchlist.html
    - history.html
    - favorites.html
    - categories.html
    - admin/
        - dashboard.html
        - movies.html
        - users.html
        - uploads.html

        - reports.html
        - settings.html
- css/
    - style.css
    - responsive.css
    - auth.css
    - movies.css
    - player.css
    - dashboard.css
    - admin.css
- js/
    - main.js
    - api.js
    - auth.js
    - navigation.js
    - search.js
    - movies.js
    - movie-details.js
    - player.js
    - profile.js
    - watchlist.js
    - history.js
    - favorites.js
    - notifications.js
    - admin.js
- assets/
    - images/
    - icons/
    - logos/

- fonts/

README.md

8. HOMEPAGE REQUIREMENTS

The homepage is the first major interface users will see after logging in.

8.1 Homepage sections

A. Navigation bar

Contains:

- FILMORA logo.
- Home navigation.
- Movies navigation.
- Genre navigation.
- Search input.
- Profile menu.

B. Hero section

Displays a featured movie.

Information:

- Featured movie backdrop.
- Movie title.
- Short description.
- Release year.
- Genre.
- Movie rating.
- Watch Now button.
- More Information button.

C. Trending movies

Displays popular movie cards.

D. Recently added movies

Displays newly published movies.

E. Popular categories

Displays collections such as:

- Action.
- Adventure.
- Comedy.
- Drama.
- Horror.
- Romance.
- Science Fiction.
- Thriller.

F. Recommended movies

Displays movies selected through recommendation logic.

G. Footer

Contains:

- About FILMORA.
- Contact information.
- Privacy policy.
- Terms of service.
- Copyright information.

9. REGISTRATION PAGE

The registration page allows new users to create FILMORA accounts.

9.1 Required fields

Field	**Requirement**
Full name	Required
Username	Required
Email	Required
Password	Required
Confirm password	Required

Optional field:

- Profile picture.

9.2 Registration functionality

- Input validation.
- Password visibility toggle.
- Password confirmation.
- Email format validation.
- Password strength feedback.
- Registration error messages.
- Registration success message.
- Link to login page.

The backend will handle account creation and password hashing.

10. LOGIN PAGE

The login page allows registered users to access their accounts.

Required fields:

- Email address.
- Password.

Additional features:

- Remember-me option where supported by secure session design.
- Password visibility toggle.
- Forgot password link.
- Login error messages.
- Account status messages.
- Redirect after successful login.

11. PASSWORD RECOVERY PAGE

The page will provide:

- Email address input.
- Submit reset request.
- Confirmation message.
- New password form.

- Confirm new password field.

Password recovery tokens will be validated by the backend.

12. MOVIE DISCOVERY PAGE

Displays available movies in an organized layout.

Features:

- Movie grid.
- Movie posters.
- Movie titles.
- Release years.
- Ratings.
- Genre information.
- Pagination or load-more functionality.
- Sorting options.
- Genre filtering.

13. MOVIE DETAILS PAGE

Each movie will have a dedicated information page.

13.1 Movie information

- Movie poster.
- Movie backdrop.
- Movie title.
- Description.
- Release date.
- Duration.
- Genre.
- Language.
- Age rating.
- Movie rating.
- Director.
- Cast.

- Available subtitles.
- Available audio tracks.

13.2 Available actions

- Watch movie.
- Add to watchlist.
- Add to favorites.
- Submit a review.
- View movie trailer.
- Discover similar movies.

14. MOVIE STREAMING PLAYER

The streaming player is a central feature of FILMORA.

14.1 Player requirements

- Play and pause.
- Volume control.
- Full-screen mode.
- Playback progress.
- Seek forward and backward.
- Playback speed selection where supported.
- Subtitle selection.
- Audio track selection.
- Video quality selection where supported.
- Resume watching.
- Loading indicator.
- Playback error messages.

14.2 Streaming technology

The frontend will use:

- HTML5 Video API.
- HLS.js where browser support requires it.
- HLS streaming manifests.

- Authorized video access from the backend.

The interface must support adaptive video quality when the streaming source provides multiple renditions.

The player must not expose private storage credentials.

15. MOVIE SEARCH FUNCTIONALITY

Users must be able to search movies by:

- Movie title.
- Genre.
- Release year.
- Language.
- Cast members.

Search features:

- Search input.
- Search results.
- Empty result message.
- Loading indicator.
- Search suggestions where supported.
- Clear search button.

Search requests will be processed by the backend API.

16. GENRE AND CATEGORY PAGES

Each genre will have a dedicated browsing interface.

Examples:

- Action Movies.
- Horror Movies.
- Comedy Movies.
- Romance Movies.
- Science Fiction Movies.
- Adventure Movies.

Users can browse movies belonging to multiple genres.

17. USER DASHBOARD

The user dashboard provides access to personal account features.

Dashboard sections:

- Profile overview.
- Recently watched movies.
- Continue Watching.
- My Watchlist.
- My Favorites.
- Account settings.
- Notifications.

18. USER PROFILE PAGE

The profile page allows users to:

- View their account information.
- Edit their full name.
- Change profile picture.
- Update personal preferences.
- Change password.
- Manage account sessions.
- Request account deletion.

19. WATCHLIST AND FAVORITES

19.1 Watchlist

The watchlist page displays movies saved by the user.

Features:

- Movie poster.
- Movie title.
- Release year.
- Remove from watchlist.
- Open movie details.
- Watch movie.

19.2 Favorites

Displays movies marked as favorites.

Users can add or remove movies from this collection.

20. WATCH HISTORY

The watch history page displays previously watched movies.

Features:

- Movie poster.
- Movie title.
- Date last watched.
- Viewing progress.
- Continue Watching button.
- Remove history entry.

21. REVIEWS AND RATINGS

Users will be able to:

- Rate movies from 1 to 5 stars.
- Submit written reviews.
- Edit their own reviews where permitted.
- Delete their own reviews.
- View reviews from other users.

The interface will display average movie ratings.

22. NOTIFICATION INTERFACE

The notification interface will display:

- New movie announcements.
- Account notifications.
- Platform announcements.
- Important security notifications.

Users will be able to mark notifications as read.

23. ADMIN DASHBOARD

The administration interface will be separate from the normal user dashboard.

23.1 Dashboard overview

Display:

- Total registered users.
- Total published movies.
- Total movie views.
- Recently added movies.
- Recent user registrations.
- Pending content reports.
- Video processing status.

23.2 Admin navigation

- Dashboard.
- Movies.
- Add Movie.
- Genres.
- Categories.
- Users.
- Media Uploads.
- Reviews.
- Reports.
- Notifications.
- Settings.
- Logout.

Only authorized administrators will access these pages.

24. MOVIE MANAGEMENT INTERFACE

The admin movie management page will provide:

- Movie listing.
- Add movie form.
- Edit movie form.
- Movie deletion or archiving.

- Poster upload.
- Backdrop upload.
- Video upload.
- Genre assignment.
- Cast assignment.
- Publishing controls.
- Content rights information.

25. USER MANAGEMENT INTERFACE

Administrators will be able to:

- View registered users.
- Search users.
- View account details.
- Change permitted account statuses.
- Manage user roles.
- Review reported accounts.
- View relevant account activity.

26. RESPONSIVE DESIGN REQUIREMENTS

FILMORA must work across different screen sizes.

26.1 Desktop

- Full navigation bar.
- Multi-column movie grids.
- Large featured movie banners.
- Full-width movie player.

26.2 Tablet

- Adapted navigation.
- Smaller movie grids.
- Responsive movie information layouts.

26.3 Mobile

- Compact navigation.

- Mobile-friendly search.
- Two-column or single-column movie grids.
- Touch-friendly controls.
- Responsive video player.
- Readable forms and text.

27. FRONTEND SECURITY REQUIREMENTS

The frontend must:

- Avoid storing passwords.
- Avoid exposing database credentials.
- Avoid exposing private storage keys.
- Use HTTPS in production.
- Communicate through authenticated backend APIs.
- Handle session expiration.
- Validate user input.
- Avoid unsafe HTML injection.
- Display safe error messages.
- Respect backend authorization decisions.

Frontend validation does not replace backend validation.

28. API INTEGRATION REQUIREMENTS

The frontend will communicate with the backend using the Fetch API.

Examples:

- Registration requests.
- Login requests.
- Movie retrieval.
- Movie searches.
- Movie details.
- Watchlist updates.
- Review submissions.
- Viewing progress updates.

- Streaming authorization.

API responses will be handled through reusable JavaScript functions.

29. ACCESSIBILITY REQUIREMENTS

FILMORA should support:

- Keyboard navigation.
- Clear focus indicators.
- Appropriate color contrast.
- Form labels.
- Accessible buttons.
- Image alternative text.
- Video captions where available.
- Readable typography.

30. FRONTEND PERFORMANCE

The frontend should implement:

- Image lazy loading.
- Optimized image sizes.
- Efficient JavaScript.
- Minimized unnecessary requests.
- Pagination.
- Responsive images.
- Reusable CSS.
- Efficient movie rendering.

31. TESTING REQUIREMENTS

The frontend must undergo:

1. Homepage testing.
2. Registration testing.
3. Login testing.
4. Navigation testing.
5. Search testing.

6. Movie details testing.
7. Streaming player testing.
8. Watchlist testing.
9. Profile testing.
10. Admin interface testing.
11. Mobile responsiveness testing.
12. Accessibility testing.
13. Browser compatibility testing.

32. FRONTEND DEVELOPMENT PHASES

Phase	Development
1	Create frontend project structure
2	Configure global styles
3	Develop navigation and footer
4	Develop registration page
5	Develop login page
6	Develop homepage
7	Develop movie discovery page
8	Develop movie details page
9	Develop streaming player
10	Develop search and filtering
11	Develop user dashboard
12	Develop profile and settings
13	Develop watchlist and favorites
14	Develop watch history
15	Develop reviews and ratings
16	Develop notifications

Phase Development

17 Develop admin dashboard

18 Integrate backend APIs

19 Perform responsive testing

20 Final frontend optimization

FINAL FRONTEND DESIGN PRINCIPLE

FILMORA will provide a modern, responsive and cinematic user interface using HTML5, CSS3 and JavaScript.

The frontend will communicate with the backend through secure REST APIs and provide a consistent experience across desktop, tablet and mobile devices.

The design will prioritize ease of navigation, movie discovery, full-length video playback and user account management.

DOCUMENT STATUS: PROPOSED — AWAITING APPROVAL

FILMORA — YOUR WORLD OF MOVIES.